

Cost function

$$\frac{1}{2m} \sum_{i=1}^m (wx^i + b - y^i)^2$$

$$\frac{d}{dw} \frac{1}{2m} \left[(wx_1 + b - y_1)^2 + (wx_2 + b - y_2)^2 \right]$$

$$\frac{d}{dw} \frac{1}{2m} \left[2[(wx_1 + b - y_1)] + [(wx_2 + b - y_2)] \right]$$

$$\frac{1}{2m} \left[2(wx_1 + b - y_1) \cdot x_1 + (wx_2 + b - y_2) \cdot x_2 \right]$$

$$\frac{\partial}{\partial w} \frac{1}{2m} \left(\sum_{i=1}^m (wx^i + b - y^i)^2 \right) =$$

$$\frac{1}{2m} \left(\sum_{i=1}^m (wx^i + b - y^i) \cdot x^i \right)$$

$$\frac{1}{m} \sum_{i=1}^m (wx^i + b - y^i) \cdot x^i$$

$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^m (wx^i + b - y^i)$$

$$\frac{d}{dx} (2x + 3)$$

$$\frac{d}{dx}$$

$$(2x)^2$$

$$2(2x)$$

Project

feature/target

constant

Cost function

↑

Changing w/b

gb

Cost "Function" returns with w and b as parameters

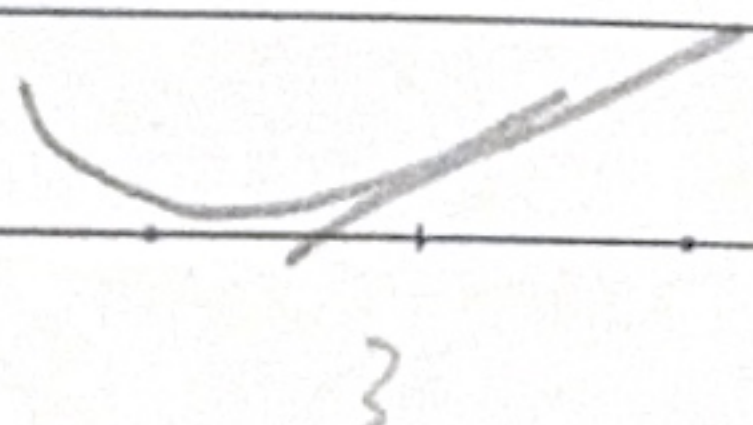
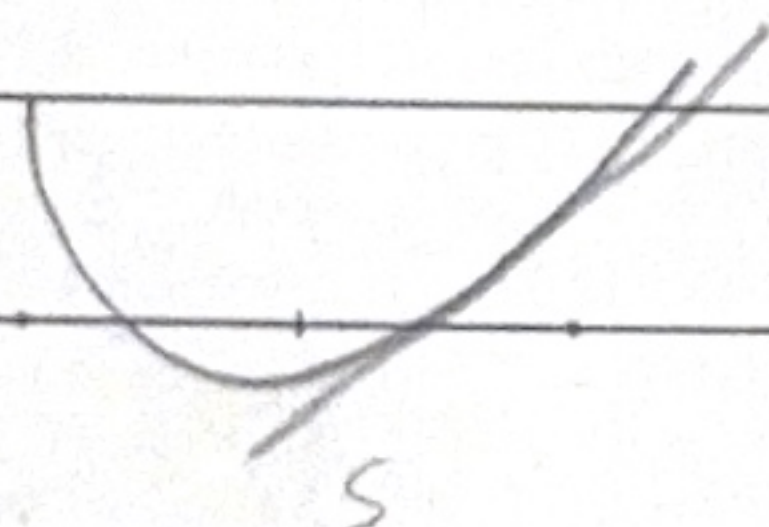
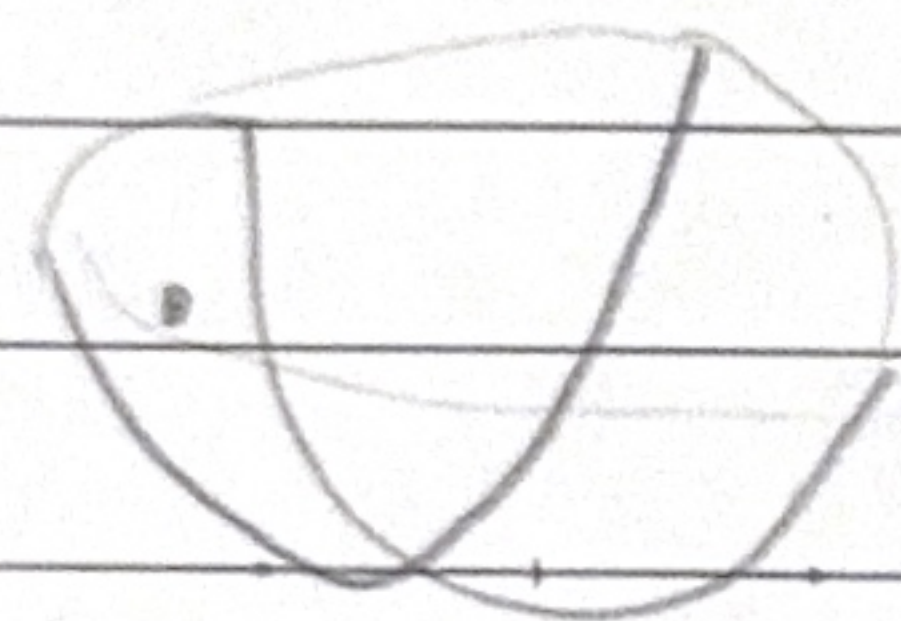
$$\frac{d}{dw} J(w, b)$$

↑ points w and b

Cost:

$$\frac{1}{2m} \sum_{i=1}^m (\vec{w} \cdot \vec{x}_i + b - y_i)^2$$

$$\frac{\partial}{\partial \vec{w}} \frac{1}{2m} \sum_{i=1}^m (\vec{w} \cdot \vec{x}_i + b - y_i)^2 = \vec{x}_i$$



$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 3 \\ 2 \end{bmatrix} = \text{predicted} - y_i$$

Featuring

x	y	z
1	3	7
5	6	8
3	9	2

A $\rightarrow$ dep variable

$\vec{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$ $N = m \times n$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 5 & 3 \\ 3 & 6 & 9 \\ 4 & 8 & 2 \end{bmatrix} = \text{Error terms} = E$$

include for bias

errors for each sample

$$E = \max | \quad |$$

$$W_i := W - \alpha \frac{\partial}{\partial W_i}$$

add a 1 column to matrix for bias

X matrix is all data point features

$$E^T \cdot X = \text{Derivative array}$$

$$\max = \text{der}$$

Scale the array by α

$$\text{Weight array} = \text{weight} + \text{array} - \text{Per array}$$

until stop